Sahaj Bhadja

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Summary

MSCS candidate (May 2026) with professional internship and production experience building scalable **Python**, **Java**, and **JavaScript** systems. Strong in OOP, automated testing, and **AI-assisted development** (Cursor, Copilot) for high-quality cloud software delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Database Systems, Software Engineering, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Structures

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Architected and implemented **Python/FastAPI services** for 500+ users on Azure, reducing latency from **1.2s to 350ms** through OOP design, SQL optimization, and telemetry-driven performance improvements.
- Designed automated validation and integration tests for 5K+ monthly jobs with CI/CD releases, improving code coverage metrics and production reliability to 99%+ in a collaborative engineering team.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built **full-stack features** with **Python, React, and REST APIs** handling 10K+ daily requests, partnering with PMs and engineers on design, implementation, testing, and customer-facing releases.
- Used **LLM-assisted tooling** to accelerate development while reviewing generated code for correctness, security, and maintainability in fast-paced product delivery.

USC AD Labs **Los Angeles, CA**
Software Engineering Assistant *January 2025 – May 2025*

- Developed **Python pipelines** and reproducible test cases across 100K+ records, supporting functional validation, regression analysis, and operational excellence in research software.
- Delivered technical documentation and test strategies that improved team alignment and engineering quality as features moved toward production readiness.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Implemented **Java/Python microservices** and AWS integrations (Glue, S3), improving API response time by **30%** while building scalable, maintainable object-oriented backend components.
- Maintained **20+ PyTest/Mockito test suites** in Jenkins CI/CD, increasing automation coverage and release confidence by **68%** across cross-functional Agile teams.

Aasma Technology Solutions **India**
Software Engineer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **JavaScript/Node.js and MERN** modules with WebSockets for 2M+ record syncs, applying OOP patterns and REST integration design for scalable customer workflows.
- Automated build/test pipelines with Docker and Jenkins, cutting release cycles by **50%** and supporting stable deployments across dev, test, and production environments.

Skills

Languages: **Python**, **Java**, **JavaScript/TypeScript**, SQL, HTML/CSS, C++
Engineering: OOP design, REST APIs, data structures/algorithms, scalable service architecture
Testing: Unit/functional/integration automation, PyTest, Mockito, test strategy design, CI metrics
Tools: Git/GitHub, AWS/Azure, Docker, Jenkins; **Cursor**, Copilot, LLM code review workflows

Projects

PiXMod Live | *Next.js, JavaScript, MongoDB, SQL*

- Built a scalable **full-stack SaaS product** with billing integrations, relational/NoSQL persistence, and iterative feature delivery used by real customers.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Designed and implemented a **Java + TypeScript** application with automated tests and multi-database integrations, demonstrating OOP and production-style software craftsmanship.

Linguistix Code | *Python, AWS, CI/CD, Automated Testing*

- Deployed a **cloud-native REST service** with CI/CD validation gates and containerized releases, emphasizing reliability, observability, and automated quality checks.